ECO344 - INTERNATIONAL ECONOMICS

MODULE ONE: ECONOMY AND GLOBAL TRADE

UNIT 1: GLOBAL ECONOMY

1.0 Introduction

International economics studies the exchange of goods, services and productive factors across national boundaries, and the effects of those exchanges on national welfare. Its trade-theory branch explains why nations specialise and how the gains are shared; its monetary branch considers exchange rates and international transmission of unemployment and inflation. Trade is not simply buying goods from abroad: policy choices, mobility restrictions and differences in resources all affect its consequences.

3.1 Perspective on the Theory of International Trade

The central questions are why trade occurs, what a country exports and imports, how commodity and factor prices respond, and who gains or loses. Within countries, firms and consumers pursue their own interests, but governments influence cross-border trade through tariffs, quotas and controls on factors. Labour is commonly treated as relatively mobile within a country but less mobile across national borders. Trade theory usually begins with full employment and aggregate trade balance, whereas international monetary analysis permits macroeconomic fluctuations and studies exchange-rate questions. A country need not be more productive in every industry to gain: differences in opportunity cost create scope for specialisation.

This level of decision-making is one feature that distinguishes the study of international trade froin the study of traditional economics. International economics can be divided conveniently into two parts: real analysis or trade theory, and monetary analysis or international finance. Because its focus is equilibrium determination of real trade flows and welfare, trade theory generally analyzes barter exchange expressed in terms of a numeraire good. Monetary analysis, on the other hand, is concerned with such issues as the determination of exchange rates and the international transmission of unemployment and inflation. One question naturally arises: why is it necessary to distinguish trade between nations from trade between regions, and even from trade between individual consumers?

3.2 Importance of International Trade

Eight practical benefits are: a wider choice of goods; better allocation and use of scarce resources; more efficient production; job opportunities as export markets expand; access to products at lower cost; reduced fluctuations through wider markets; outlets for surplus production and foreign-exchange earnings; and goodwill arising from commercial relations. Countries specialise in activities suited to their resources and use imports to meet demands that domestic production cannot satisfy cheaply. The gain is not automatic for every industry or worker, but the potential economy-wide improvement is central to trade analysis.

Nations with strong international trade have become prosperous and have the power to control the world economy. Global trade can become one of the major contributors to the reduction of

poverty. Several benefits that can be identified with reference to international trade are as follows: 1) Greater Variety of Goods Available for Consumption: International trade brings in different varieties of a particular product from different destinations. This gives consumers a wider array of choices which will not only improve their quality of life but as a whole it will help the country grow. 2) Efficient Allocation and Better Utilization of Resources: Efficient allocation and better utilization of resources since countries tend to produce goods in which they have a comparative advantage.

PAST QUESTION - 2025_1 Q1(a); 2022_2 Q2(a)

Question: Discuss the importance of international trade, especially for Nigeria.

Answer: International trade widens access to goods unavailable or costly to produce domestically; encourages specialisation where comparative costs are lower; promotes efficiency by increasing competition; allows the sale of surplus output; expands possible markets, production and employment; makes machinery and technical know-how available; and supports foreign-exchange receipts. It can encourage investment in transport and communication and promote cooperation. For Nigeria, the principles mean export earnings can finance imports of equipment and inputs, while firms may reach buyers beyond the domestic market. These benefits must be weighed against import dependence and pressure on local industries.

3.3 Advantages and Disadvantages of International Trade

Advantages include specialisation, economies of large-scale production, access to technology and equipment, price stability, improved transport and communication, and international cooperation. Imports can supplement food and essential goods after natural disasters. Disadvantages include exposure of infant domestic industries to stronger competitors, possible economic or political dependence, excessive extraction of natural resources, undesirable imports, shortages of essential exports at home, vulnerability during hostilities, and rivalries between trading nations. When answering an essay, separate the economy-wide potential gains from the risks faced by particular groups.

The benefits and costs operate at different levels. A firm may gain access to imported machinery and cheaper inputs, while another firm may lose sales to competitive imports. For a country depending on one export commodity, changes in international prices can quickly affect government revenue and the supply of foreign exchange. Trade also creates a need for transport networks, customs administration, ports and dependable communications. During a crop failure or other domestic shortage, imports can prevent a temporary supply problem from becoming a more serious crisis. However, advantages such as low import prices should not be confused with proof that every domestic producer benefits.

PAST QUESTION - 2025_1 Q2(b); 2022_1 Q5

Question: Explain important adverse effects of international trade on developing economies.

Answer: A risk is that newly established domestic firms may not survive competition from large foreign producers. Economies may become dependent on imported technology or export receipts from a narrow range of commodities. Over-exporting can deplete resources or cause home shortages. It also discusses dual economies, limited gains for poorer countries, imitation of foreign consumption patterns (the demonstration effect), and deterioration of the terms of trade for some primary products. The key distinction is between potential aggregate gains and an unequal distribution of those gains.

3.4 Trade, Growth and Economic Interrelatedness

Trade integration means national economies become more dependent on exports, imports and flows of investment. A comparison can be made between growth in world merchandise exports and output, noting that trade sometimes grew faster than total production. The export-to-GDP ratio gives one useful indication of a country's openness, although it should be interpreted alongside the economy's size and structure. Causation is not automatic: expanding trade can support production and growth, while economic growth can itself increase the ability and desire to trade. International transactions link prices, employment and policy choices across borders.

At the same time, global growth in real output, measured by gross domestic product (GDP) in each country, averaged 6.1 percent per year, also high by historical standards. This trend continued after 1979, though economic activity grew at markedly slower rates. This increasing interrelatedness among countries may be observed for specific nations as well. Table 1.1 lists a selected set of countries at different levels of economic development. Clearly there is wide variation in international living standards, as measured by per-capita GNP. While there are problems in constructing such measures, it appears that there may be as much as a one hundred-fold difference in per- capita incomes between the poorest and wealthiest countries of the world.

PAST QUESTION - 2022_1 Q1(a); 2025_2 Q3(a)

Question: Explain the link between international trade, economic growth and integration.

Answer: As exports and imports expand, producers gain access to larger markets, consumers to a broader range of goods and economies to specialised inputs. Export earnings can finance capital goods, while international competition may stimulate efficient production. Growth also feeds trade because higher incomes, productivity and capacity raise demand for imports and the supply of exports. Integration is therefore a two-way relationship, not a claim that all countries or groups benefit equally. An important issue is growth of trade relative to global output and uses trade-to-GDP measures to illustrate interrelatedness.

3.5 Trade and National Characteristics

Countries differ in factor endowments, technology, population, income, institutions and preferences. These differences influence comparative costs, trade structure and exposure to international competition. A densely populated, labour-abundant economy may face a different production pattern from one with substantial capital or natural resources. Another consideration is the importance of foreign direct investment and international labour movements. Countries therefore differ not only in what they trade but also in how much of their production and income depends on foreign markets.

For example, much of Malaysia's exports are processed through Singapore to their ultimate destinations elsewhere. Note also that Canada has long had a high proportion of its GDP devoted to exports, with a slight rise to nearly one quarter by 199. For example, in Canada exports now tend to comprise a larger component of national demand than investment. GDP in 1991 was produced for export, a figure that amounted to some $398 billion. United States is such a large country that relatively little of its output needs to be produced for the foreign sector. The European Community (EC) provides a good example of a set of countries that are intimately interrelated through international transactions.

PAST QUESTION - 2024_1 Q1(a)

Question: Briefly appraise foreign direct investment in international trade.

Answer: Foreign direct investment refers to investment across borders that establishes or extends a lasting productive business presence. In this setting, it forms part of the growth of multinational enterprises whose production and marketing operations span several countries. It can connect host countries with capital, production capacity and international distribution, while also raising questions about the location of jobs, environmental regulation and the allocation of gains.

3.6 The Sectoral Structure of Trade

The sectoral structure describes the composition of internationally traded production by industry or sector, such as agriculture, extraction, manufactures and services. A country exporting raw materials but importing advanced manufactures has a different pattern from one exporting technologically sophisticated products. Such patterns reflect available resources, technology and demand. Trade can be inter-industry, where unlike sectors exchange goods, or involve related manufactures; in either case identify which sectors earn export income and which rely on imports.

Looking at two major domestic sectors, in 1990 the United States exported over 45 percent of its agricultural production and imported over 43 percent of its consumption of motor vehicles and automobile parts. The calculations are designed to reveal a rough measure of comparative advantage by sector in each country. Some brief comments about the sectors in this table are in order. The United States is noteworthy in that although it is one of the largest petroleum producers in the world, it remains a major importer because of its huge demand for energy. For example, Germany, the United Kingdom, the United States, and Switzerland are all major exporters of chemicals, while Canada and Japan tend to Import them, as do the developing countries.

PAST QUESTION - 2024_1 Q1(b)

Question: What is the sectoral structure of trade?

Answer: It is the way a country's foreign trade is distributed among economic sectors and products. Examples are primary products, agricultural produce, manufactures and services. Studying the structure reveals which industries dominate exports and which depend on imports, and helps explain how endowments, industrial development and technology shape a nation's trade relationships.

Unit Quick Revision

International trade crosses national borders; trade theory studies real quantities and welfare while international finance considers monetary adjustments. Benefits include variety, efficiency and larger markets; risks include import dependence and unequal outcomes.

UNIT 2: SUPPLY AND PRODUCTION POSSIBILITIES

1.0 Introduction

Production theory explains what an economy is capable of producing before trade takes place. Labour and capital can be combined in different proportions; technology determines the quantity obtained. The production function, returns to scale, isoquants and cost constraint are tools for finding efficient production.

3.1 Production Functions

A production function represents output by X = F(K,L), where K is capital, L is labour and F represents technology. An isoquant joins combinations of capital and labour that produce the same output. With capital fixed, the total product of labour may rise as more labour is added, but often at a falling rate. Marginal product of labour is the extra output from a small rise in labour input: MPL = change in X / change in L, with capital held fixed. Diminishing marginal returns are compatible with constant returns to scale because the first changes one input and the second changes all inputs together.

An isoquant is drawn for one specified output level. Any combination of K and L on it produces that level; higher isoquants represent larger output. The marginal product of a factor depends on how much of the other factor is available. For example, extra labour on a fixed quantity of machinery can initially increase output, but each additional worker may contribute less than the previous worker. Giving the workers more machinery can shift the total product curve upwards. Keep the distinction clear: moving along a fixed isoquant changes the factor combination, whereas moving to a higher isoquant increases output.

PAST QUESTION - 2022_1 Q1(b)

Question: Use an equation to describe a production function.

Answer: A production function expresses the maximum output available from a given technology and inputs. In the two-factor model, X = F(K,L), where X is quantity produced, K denotes capital services, L denotes labour services and F captures the technique of production. For example, Q = 0.5KL describes output as a function of both factors. Holding capital constant and increasing labour helps identify labour's marginal product; increasing both factors together allows returns-to-scale analysis.

3.2 Returns to Scale

Returns to scale concern a simultaneous proportional rise in every productive input. If F(mK,mL) = mF(K,L), output rises in the same proportion and returns are constant. If F(mK,mL) exceeds mF(K,L), returns are increasing; if it is less, returns are decreasing, with m greater than one. For a Cobb-Douglas function Q = A K^a L^b, compare a+b with one: equal gives constant, greater gives increasing and smaller gives decreasing. This is different from the law of diminishing marginal product, which holds one or more other factors constant.

The function X = F(K, L) is said to be homogeneous of degree k if λ X = F(λK, λL). If the function is, homogeneous and if k is equal to 1, then the output will also double. Consider first the isoquant X0, where it has been assumed that the level of output is equal to 10. One such point, A, shows that 10 units of output can be produced using 6 units of labor and 7 units of capital. Since the capital/labor space is “full” of isoquants, there must be one that passes through point B.

PAST QUESTION - 2021_2 Q1(c,d); 2022_2 Q1(a,b)

Question: Explain constant, increasing and decreasing returns to scale and classify Q = 0.5KL, Q = 2K+3L, and Q = K^0.3 L^0.2.

Answer: A constant-returns production function doubles output when every input is doubled; increasing returns more than double it; decreasing returns less than double it. For Q = 0.5KL, multiplying K and L by m multiplies output by m^2, so returns are increasing. For Q = 2K + 3L, output becomes mQ, so returns are constant. For Q = K^0.3 L^0.2, output becomes m^0.5 Q, so returns are decreasing. The sum of powers is useful for Cobb-Douglas functions; it is not a shortcut for every functional form.

3.3 Equilibrium for a Single Producer

A producer either maximises output for a fixed expenditure or minimises expenditure for a fixed output. If the input budget is C0 and prices are wage w and capital rental r, the isocost condition is C0 = wL + rK. Optimal input choice occurs when the highest attainable isoquant touches the isocost line. Therefore the marginal rate of technical substitution equals the input-price ratio: MPL/MPK = w/r (where both marginal products exist and an interior solution applies). The tangency shows why picking a point on a lower isoquant wastes the productive possibilities of the given budget.

The behavioral assumptions for an individual producer can be stated in either of two entirely equivalent ways: The producer can be thought of as maximizing output subject to a cost constraint or as minimizing costs subject to a production constraint. It is assumed that the wage rate, w, and the rental on capital equipment, r, are known to the producers. It is further assumed that each individual producer is too small to have any influence on the price of his or her inputs, so that w and r can be treated as constants regardless of the level of output.

PSALMDATA

PAST QUESTION - 2022_2 Q1(c)

Question: Given Q = 0.5LK - 0.1L² - 0.05K², capital costs ₦25, labour ₦20, and the total budget is ₦500, obtain the maximum-output input choice.

Answer: Use 25K + 20L = 500, hence K = 20 - 0.8L. Substituting into production gives Q(L) = 0.5L(20-0.8L) - 0.1L² - 0.05(20-0.8L)² = -20 + 11.6L - 0.532L². Setting the derivative 11.6 - 1.064L equal to zero gives L approximately 10.90 units and K approximately 11.28 units. The second derivative is -1.064, so this is a maximum. Substitution yields maximum output approximately 43.23 units. Fractional factors are permissible in this continuous production model.

PAST QUESTION - 2024_1 Q2

Question: Explain tangency and convexity conditions for maximising production value.

Answer: At an interior efficient point, the producer chooses the feasible input or output mix at which the supporting price line just touches the relevant production frontier or the budget line touches the highest attainable isoquant. Equality of the relevant slopes is the tangency condition. Convex technology sets, or a concave production possibility frontier facing a supporting price line, help ensure that this tangency is a global maximum rather than a minimum or an unstable intersection. When the frontier is not convex, a corner or specialised solution may instead maximise value.

3.4 Objectives of Trade Policy

The supply model also supports trade-policy questions. Because countries differ in technologies and factor supplies, efficient production creates different opportunity costs and therefore possible trading patterns. Governments intervene through tariffs, quotas or other measures for purposes that may include protecting domestic production or influencing imports and exports. Such interventions have economic costs and benefits that must later be assessed by comparing producer, consumer and world prices.

In practice, despite advice of economists, every nation protects its own domestic production to some extent from foreign competition. It is important to know that behind these barriers to trade are special interest groups whose jobs and incomes are threatened. To this effect, they cry to the government for protectionism, (Appleyard and Field, 1998). Protectionism refers to government use of embargoes, tariffs, quotas and other restrictions to protect domestic producers from foreign competitors, (Tucker, 2011). The instruments of trade policy provide an overview of the various instruments of trade policy available to government policy makers, (Appleyard and Field 1998) Over the years, international trade has been emphasized as a strategy of fostering economic growth by both developed and developed countries.

Unit Quick Revision

Production function: X = F(K,L). Returns to scale compare changes in all factors; diminishing marginal returns hold other factors fixed. Producer equilibrium: C = wL + rK and MPL/MPK = w/r at an interior tangency.

UNIT 3: GOODS AND FACTOR MODEL

1.0 Introduction

A two-good, two-factor economy links the allocation of labour and capital to the quantities of goods X and Y that can be produced. The same resources cannot be used twice: producing more of one good usually means moving inputs away from the other. This trade-off is shown by the production possibility frontier.

3.1 The Two-Good, Two-Factor Model

Let both goods X and Y use labour and capital. Total national endowments are fixed, so Lx + Ly = L and Kx + Ky = K. Efficient allocations are those where a reallocation cannot increase output of one good without reducing the other. An Edgeworth-Bowley box can display feasible factor allocations for both industries at once. At an efficient interior point the slope of the isoquant is the same in both sectors, meaning the marginal rate of technical substitution is equal. Different factor intensities help explain curvature of the aggregate frontier.

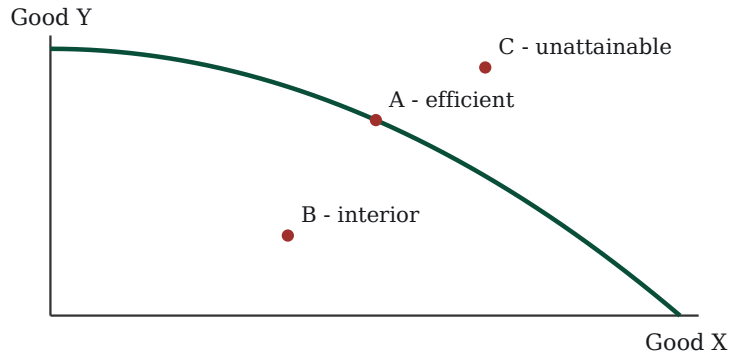
These production functions are assumed to be homogeneous of the first degree and are assumed to be increasing functions of both inputs. It is further assumed that positive outputs imply positive inputs of both factors. The economy is assumed to have fixed total supplies of both capital and labor, and these two constraints are represented by Eqs. : K = Kx + Kx L = Ly + Ly As well as showing the allocation of the two factors between the two production processes, the equality sign in these two equations implies that; these two processes use all the available K and L.

3.2 The Shape of the Production Possibility Frontier

The PPF is the outer limit of feasible combinations of X and Y when resources and technology are fixed. Points on it are productively efficient; points inside represent possible production that is not fully efficient; points beyond it are unattainable with present resources. As production moves toward more X, labour and capital may be withdrawn from Y at rising opportunity cost, making the frontier concave to the origin in the conventional depiction. Its slope in absolute value is the marginal rate of transformation, MRT, measuring units of Y forgone for an extra unit of X.

The first, which we might call engineering efficiency, implies simply that for either of the production functions and for any bundle of inputs, output is as large as it could possibly be. Suppose that all the labor and all the capital were allocated to the production of commodity Y, so that in Eqs. , Kv and Ly are replaced by K and L. To obtain some idea of where this curve might be, construct the straight-line joining Y and X and consider whether points on this line arc possible production points. All other points on the line YX could be generated in a similar fashion, so all point on this line are feasible production points.

PSALMDATA



PPF: feasible output frontier

Outside: not currently feasible

Inside: less than full efficiency

Slope measures opportunity cost

PAST QUESTION - 2022_1 Q2; 2024_1 Q3(b); 2025_2 Q1(a)

Question: Define and illustrate the production possibility frontier.

Answer: The production possibility frontier (PPF) is the boundary of output combinations of two goods attainable from existing factor endowments and technology when resources are efficiently used. Put good X on the horizontal axis and good Y on the vertical axis. Draw a downward-sloping frontier bowed away from the origin to illustrate rising opportunity cost. A point on the frontier is efficient; an interior point indicates unused or misallocated productive potential; an exterior point is unattainable without additional resources or technology. Its slope is the marginal rate of transformation between the two goods.

PAST QUESTION - 2021_2 Q3(c)

Question: State three conditions determining the PPF's position and shape.

Answer: The no-trade analysis identifies (1) available endowments of primary factors, including labour and capital; (2) production technologies and productivity in the two sectors; and (3) returns-to-scale properties of the production functions. Greater resources or improved technology can shift the frontier outward; the technologies and relative factor intensities affect its curvature and opportunity costs.

3.3 Competitive Equilibrium

Competition causes firms to allocate factors so that their value contribution accords with factor prices. For producers, the value-maximising mix of goods is reached where the relative commodity price Px/Py equals MRT, provided the optimum is an interior tangency. In factor markets, the price of labour is the wage, while the price of capital services is the rental rate. Goods prices and factor prices thus link the production decisions of both sectors.

It has also been shown that for the individual producer, the maximization of production subject to the cost constraint requires that the ratio of factor prices be equal to the slope of the isoquants. At point A, the isoquants X0 and Y0 are both tangent to the wage-rental w/r and hence are tangent to each other. The value of the marginal product of a Eqs. as Px = MPLY = MPKY Py MPLX MPKX factor is the price of the good times the "physical" marginal product of the factor we discussed earlier in the chapter.

3.4 Increasing Returns to Scale

With increasing returns, larger scales of operation yield more than proportional output gains. Such technologies may make the aggregate production set non-convex and lead to production concentrated in particular sectors. In this setting, increasing returns commonly raise questions about perfect competition, because a larger producer may gain an advantage. This matters for trade because scale economies can influence which goods are produced and traded, beyond differences in resources and productivity alone.

Scale economies are different from a productivity gain caused by adding just one input. If the quantity of every input rises by ten per cent and output rises by more than ten per cent, the production activity displays increasing returns to scale. Large firms may then have lower average costs than small firms, providing a reason why production becomes concentrated. Because the production frontier can be non-convex, a competitive price-line tangency is not always enough to prove that a candidate production point is the highest-valued attainable point. This qualification matters when proving welfare gains from trade under conditions other than constant returns.

3.5 The Basis of Trade

International trade can arise because countries have different production possibilities, tastes, factor endowments or technological efficiencies. A country’s comparative advantage lies in the good with the lower opportunity cost, not necessarily in the good for which its physical output per worker is absolutely highest. The production frontier identifies domestic transformation possibilities; world prices may offer a more favourable exchange rate and allow consumption beyond the domestic PPF.

The trade that occurs when residents of a country trade with residents of other countries is termed international trade. In the exchange of goods and services among different countries, foreign currencies are required. As a result, a more useful definition of international trade is the trade that cuts across national boundaries and involves the use of different currencies. The intellectual insights of Adam Smith were influential in bringing the mercantile system to an end. Smith and others at the time argued that governments’ mercantilist policies were influenced by special interest groups. In his famous book, “The wealth of Nations”, Smith argued that if a country specialized in what it produced best and freely traded those goods, then society will be better off.

PAST QUESTION - 2025_2 Q4(a); 2023_2 Q1(a)

Question: Differentiate absolute advantage from comparative advantage.

Answer: Absolute advantage exists when a country uses fewer resources or has greater productivity in producing a good than another country. Comparative advantage exists when its opportunity cost of that good is lower than elsewhere. A country may have absolute advantage in both goods, yet comparative advantage in only one if the relative productivity differences are unequal. Trade is guided by the relative costs: each country tends to specialise in the good it sacrifices less to produce and obtains other goods through exchange.

PAST QUESTION - 2025_2 Q2; 2022_1 Q3

Question: Why do specialisation and exchange permit gains from international trade?

Answer: Countries face different opportunity costs. By shifting production toward goods with lower domestic opportunity cost, each country can attain more valuable output at world prices. It then exchanges part of the specialised output for goods that would have cost more to produce at home. The opportunity to consume a bundle beyond the original domestic production frontier is the essential gain. Gains from specialisation concern more efficient production; gains from exchange concern consuming at favourable relative prices.

Unit Quick Revision

Efficient PPF points lie on the boundary; interior points waste feasible production; exterior points require more resources or technology. MRT is the opportunity-cost slope. Comparative advantage means lower opportunity cost.

UNIT 4: GENERAL EQUILIBRIUM IN A CLOSED ECONOMY

1.0 Introduction

General equilibrium combines production and consumption decisions. In a closed economy, all goods consumed must be produced domestically. Opening the economy changes the resource constraint from direct equality of each good produced and consumed to equality of the values of exports and imports at world prices.

3.1 General Equilibrium Analysis

In autarky, three conditions must hold simultaneously. First, competitive producers select output so that Px/Py = MRT, the producer optimum. Second, consumers choose their basket so that Px/Py = MRS, the consumer optimum. Third, markets clear: Xc = Xp and Yc = Yp. The equilibrium is a point where the PPF, a price line and a community indifference curve are tangent in the standard model. The notation p stands for prices, c for consumption and p subscripts on quantities for production.

The third condition is a market clearing condition: The supply and demand for each commodity must be equal; let subscript c denote consumption of a commodity and subscript p denote production of a commodity. Producers produce optimally at point A, where the slope of the production frontier is tangent to the price ratio, pa. And finally, markets clear because the production and consumption points are the same. Note also that the equilibrium at A is optimal in the sense that the economy consumes on the highest possible community indifference curve at which production is feasible (i.e., where the production point is on or interior to the production frontier XY).

PAST QUESTION - 2025_2 Q1(b); 2024_2 Q1(b)

Question: State and explain the three conditions for general equilibrium in a closed economy.

Answer: (1) Producer optimisation: the relative goods price equals the marginal rate of transformation, Px/Py = MRT, so output value is maximised on the PPF. (2) Consumer optimisation: the relative goods price equals the marginal rate of substitution, Px/Py = MRS, so the consumer chooses the best affordable basket. (3) Market clearing: production equals consumption for each good, Xp = Xc and Yp = Yc. Together these conditions locate the autarky equilibrium.

3.2 General Equilibrium in an Open (Trading) Economy

With trade, relative world prices p* = Px*/Py* generally differ from autarky prices. The optimisation rules remain: producers use MRT = p* and consumers use MRS = p*. The crucial replacement is the balance-of-trade equation Px*Xp + Py*Yp = Px*Xc + Py*Yc. A country can import more of one good than it produces if exports of the other good pay for it. The consumption point may therefore lie beyond its PPF even while its production point remains on the frontier.

Producers optimize by equating the marginal rate of transformation to whatever prices prevail, and, similarly, consumers optimize by equating their marginal rate of substitution to those prices. With international trade, an economy is no longer constrained to consume only what it can produce. A trading economy is able to sell some of one good at world prices and use the proceeds to buy the other commodity. If excess demand is negative, an economy is consuming less than it is producing, resulting in an export good. An alternative way of saying this is that the sum of the value of the country's excess demands must equal zero: the positive excess demand for the import good must equal the negative excess demand for the export good.

PAST QUESTION - 2021_2 Q3(a)

Question: Briefly discuss the trade-balance condition.

Answer: For an open economy, total spending on traded consumption must equal the value of traded production at world prices: Px*Xp + Py*Yp = Px*Xc + Py*Yc. Equivalently, the value of excess import demand is offset by export supply: Px*(Xc-Xp) + Py*(Yc-Yp) = 0. If a country imports X, its excess demand for X is positive; it must have a compensating negative excess demand, or exports, for Y in the two-good model.

3.3 The Excess Demand Function

Excess demand for X is Ex = Xc - Xp; for Y it is Ey = Yc - Yp. A positive value denotes an import demand, and a negative value denotes a willingness to export. Given the trade-balance condition, Px*Ex + Py*Ey = 0. A country's desired trades thus depend on world relative prices: consumers and producers adjust to those prices, changing the excess-demand quantities.

If the relative price of x y X is lower on the world markets than on the domestic market, then buying froin the low- cost source would mean importing the good. At the price ratio p* = Pa in Fig. 4.3, producers pick point Q2 and consumers pick point C2. With the price ratio greater than the autarky price ratio, the time country exports X (the relatively valuable good oh the world markets) and imports Y (the relatively cheap good on the world market). Fig. 4.4, we construct an excess demand curve for good X for the country.

3.4 Shape of Excess Demand Curves

Excess-demand curves describe the response of a country's net import or export demand to price changes. At an autarky price there is no need to trade. A change in relative prices alters domestic output decisions and consumption choices, causing net demand for one good and net supply of the other. The curve need not have a simple shape under every possible set of preferences or technologies, so the analysis concentrates on the economic link between price incentives and net international transactions.

Any price movement away from pa will elicit a response from producers and consumers. Producers will choose to move resources out of the production of X and into the production of Y. This will exacerbate any given excess demand for X at P* < Pa, leading to the negative slope of the excess demand curve. The falling price of X makes consumers willing and able to buy more of it, and again the excess demand for X grows as its price falls. Furthermore, the concavity of the production possibilities curve and the concavity of community indifference curves will combine to ensure that the excess demand function will itself be convex.

3.5 International General Equilibrium

The world market clears when one nation's export supply is matched by another's import demand at the common world price ratio. Thus global excess demand sums to zero for each traded good. The equilibrium world price reconciles the trading plans of countries and determines the composition of international exchanges. This condition differs from the closed-economy condition that each country itself produce exactly what it consumes.

General equilibrium in the world economy is then determined at an international price ratio where the excess demands of the two countries are equal and opposite. At that price, the positive excess demand (imports) of the Foreign country are equal to the negative excess demand (exports) of the Home country. The market for X clears, which is a condition for international equilibrium: Ex + E* = * P* * P* Pa * Ex -(xc – xp) E* O Ex (xc – xp) Note for future reference that the equilibrium price lies between the autarky prices of the two countries.

PAST QUESTION - 2025_1 Q1(b)

Question: Describe key conditions of international economic integration represented by the trade model.

Answer: International equilibrium is explained in terms of a common set of world relative prices, production and consumption responses to those prices, and trade balance/market clearing across countries. Producers equate MRT with the world price ratio; consumers equate MRS with the relevant price ratio; and net import demand in one country is matched by net exports elsewhere. Together they make mutually consistent international trade possible.

Unit Quick Revision

Closed economy: Px/Py = MRT = MRS and production equals consumption. Open economy: same optimisation principles, but trade balance replaces commodity-by-commodity market clearing.

MODULE TWO: THE THEORY OF GAINS FROM TRADE

UNIT 1: GAINS FROM TRADE

1.0 Introduction

Gains from trade mean the improvement in possible consumption or welfare when a country moves from autarky to trade. The central reasoning is simple: world relative prices can allow a nation to obtain goods more cheaply through exchange than through domestic production alone.

3.1 Gains from Trade

Start from the autarky point A where consumption and production coincide. When trade opens at a new relative price, production changes toward a point Q that maximises output value. Trade then permits a consumption point C on a national budget line through Q; C can be preferred to A. A country typically exports a good that is relatively cheap to produce and imports one that is relatively expensive at home. Both partners may gain even when their volumes of trade and improvements in welfare differ.

Another popular view is that if one country gains through trade, the other country must lose. In other words, while a country's total income is increased by trade, these gains may be very unevenly distributed to the point where some individuals or groups are worse off. Autarky equilibrium occurs at point A, with the economy reaching utility level Ua. Note also that if the world price ratio happens to equal the autarky price ratio, the country is no worse off. Another thing to note in Fig. 5.1 is that the direction of trade is of no particular significance.

PAST QUESTION - 2025_2 Q1(c); 2024_2 Q1(c)

Question: What is gain from specialisation?

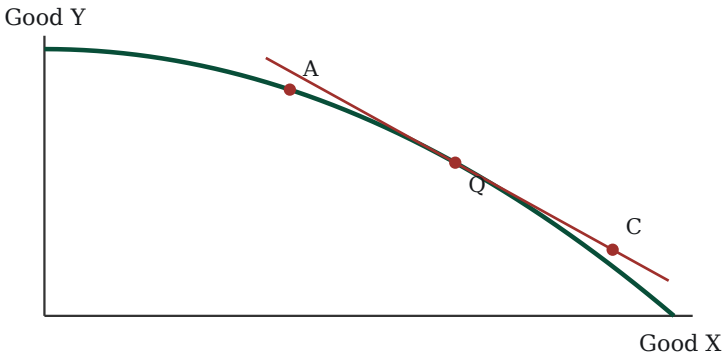
Answer: Gain from specialisation is the increase in productive opportunities from reallocating resources toward activities in which the country has comparative advantage. It occurs because a nation can produce a higher-value combination of output at world prices than under its earlier autarky allocation. It is distinct from the gain from exchange, which arises when the country trades at relative prices that allow a better consumption basket.

3.2 The Gains-from-Trade Theorem

The gain can be demonstrated by comparing the value of production at the world price ratio. If free-trade production maximises value at free-trade prices, the value of that output cannot be less than the value of the autarky output at those same prices. Balanced trade converts that production-value advantage into an affordable consumption set. Consumers can still choose the autarky bundle, but with a wider opportunity set they can obtain higher welfare. A supporting price line tangent to the PPF is a standard sufficient condition; if the frontier is non-convex, tangency may not ensure maximum value.

In Fig. 5.3, world price p* result in production at point Q for reasons discussed earlier. In particular, we see that the value of production at Q is greater than the value of production at A, the autarky equilibrium. When the value of production is maximized at free trade prices, we have the following inequality: P*xXfp + P*y Yfp > p*xXap + p*yYap In words, the value of free

trade production at free trade prices exceeds the value of autarky production at free trade prices. P*xXfc + p*yYfc > p*xXac + p*yYac P* P* This inequality states that the value of free trade consumption evaluated at free trade prices exceeds the value of autarky consumption evaluated at free trade prices.



A = autarky point

Q = production with trade

C = consumption after trade

C is outside the domestic PPF

PAST QUESTION - 2022_1 Q3; 2024_1 Q5(a)

Question: Use a graphical or verbal argument to explain the gains-from-trade theorem.

Answer: Draw a production possibility frontier with an autarky point A. At a different world price ratio, producers move to a value-maximising point Q where a world-price line supports the frontier. This line permits trade and a consumption point C that was not attainable in autarky. Because the autarky bundle remains affordable at the free-trade budget while the best available bundle is chosen, the consumer can reach greater satisfaction. Gains require attention to efficient production and the terms on which imports are exchanged for exports.

3.3 Factors Affecting Gains from Trade

The analysis lists six determinants: differences between trading partners' cost ratios; demand and supply conditions; factor availability; country size; the terms of trade; and productive efficiency. Large differences in comparative costs can create sizeable opportunities, though realised gains depend on world relative prices. A small country can often specialise more extensively; improvements in productive efficiency can expand gains. The terms of trade indicate how much imported output can be purchased with a given quantity of exports. Remember that gains to the nation do not establish equal benefits for every citizen.

These factors do not work in isolation. An improvement in technology may reduce a country's opportunity cost, while its export demand and terms of trade determine how much foreign output it can obtain in return. A small nation may enjoy access to a market much larger than its domestic economy, but a heavy concentration in one export product can expose it to changing world prices. Likewise, plentiful labour is useful only if it can be combined with appropriate capital, technology and markets. When answering an examination question, name the factor first and then explain the economic channel through which it enlarges or limits the gain.

PAST QUESTION - 2025_1 Q2(a); 2022_1 Q4; 2021_2 Q4(b)

Question: Explain five factors affecting gains from international trade.

Answer: (1) Cost-ratio differences: greater differences in opportunity costs create more room for mutually beneficial exchange. (2) Demand and supply: responsiveness influences how trade expands. (3) Factor availability: an abundance of particular resources encourages specialisation. (4) Country size: smaller markets can gain substantially from access to world production and demand. (5) Terms of trade: favourable export prices relative to import prices allow more imports for each unit exported. A further factor is productive efficiency as a sixth factor.

Unit Quick Revision

Gains from trade depend on the world price line providing consumption possibilities beyond autarky. Production at free-trade prices must maximise value in the standard proof. Remember six determinants of gain.

UNIT 2: GAIN FROM EXCHANGE

1.0 Introduction

A country can improve welfare not only by changing what it produces but also by exchanging what it has for another combination of goods. Production gains and exchange gains should be distinguished, even though they commonly arise together.

3.1 Gains from Exchange

Even if a country's production mix were fixed, the ability to exchange part of that output at favourable foreign relative prices might allow consumers to choose a more preferred bundle. This is a pure gain from exchange. In a standard diagram, hold production at the initial point but introduce a new price line; the consumer chooses the highest indifference curve available on that line. When production can also change, both exchange and specialisation contribute to total gains.

Suppose we have two individuals, Jim and Janet, and the Jim has six bottles for beer and no bags of peanuts, while Janet has five bags of peanuts but no beers. For example, Janet could give Jim three bags of peanuts for two beers. They would then move to point C in their respective diagrams, each attaining a utility level of Uf. An implicit trading price is established, in so far as three bags of peanuts are deemed to be equal in value to two beers. Beer is relatively more valuable; the price of beer in terms of peanuts is 3/2 = 1.5.

3.2 Gain from Specialisation

Specialisation uses available labour and capital more heavily in the activity with lower opportunity cost. It can raise the value of national output at world prices. Absolute advantage measures physical productivity or input use compared across countries, while comparative advantage depends on relative opportunity costs within and between countries. It is important that a country with an absolute advantage in both goods can still benefit by concentrating in the good in which its comparative advantage is greatest.

In modern society no one is self- sufficient, and indeed, most people engage in an extremely narrow range of work activities in order to earn income to buy a wide range of goods and

services. The same principle holds true for countries, although people seem to lose sight of this fact. Consider a simple example in which we have two countries, the United States and Japan that produce two goods, wheat and steel. One labour-year devoted to wheat production in the United States results in 30 tons of wheat, and so forth. Suppose we now move one worker in the United States out of steel production and into wheat production.

PAST QUESTION - 2022_2 Q4(b); 2025_2 Q1(c)

Question: Distinguish gains from exchange and gains from specialisation.

Answer: Gain from exchange is the welfare improvement from swapping goods at more favourable relative prices while holding production given. Gain from specialisation is the increase in the value or feasible quantity of output from reallocating production toward goods with comparative advantage. When trade begins, a country often changes its production point and then exchanges output at world prices, so both sources of improvement can appear together.

PAST QUESTION - 2025_2 Q4(a); 2025_1 Q3(a)

Question: Explain absolute and comparative advantage with an illustration.

Answer: Suppose one country has a greater output per worker in both commodities. It has an absolute advantage in both, but that does not settle the pattern of trade. Compare the relative unit labour requirements: if X takes fewer labour hours relative to Y at Home than it does Abroad, Home has comparative advantage in X and Abroad has comparative advantage in Y. A world exchange ratio lying between those countries' domestic opportunity costs gives each an incentive to specialise and trade.

PAST QUESTION - 2025_2 Q2; 2024_2 Q2

Question: Discuss specialisation and exchange as features of modern international trade.

Answer: Modern trade involves countries, firms and regions concentrating production in goods for which their resource use and opportunity costs are relatively favourable, then exchanging output to satisfy varied consumption needs. Specialisation can increase productive efficiency or exploit scale; exchange makes imported goods available at terms preferable to domestic production. These processes link national markets, but the results depend on prices, technology, resource endowments, demand and policy barriers.

3.3 Distribution of Gains with Heterogeneous Tastes

People differ in preferences; they may value combinations of X and Y differently, even when they face the same price ratio. This changes how much each person benefits from access to foreign trade. The analysis uses community indifference-curve reasoning to show that aggregate gains do not imply every individual enjoys an identical improvement. Distribution must therefore be studied separately from the existence of total gains.

Consider consumers who have equal incomes but different tastes for two goods. At the same world price ratio, one may gain substantially by being able to buy more of an imported good, while another may choose a basket close to the pre-trade bundle. A common market price does not imply common welfare gains because individual indifference curves differ. The economy can show an aggregate gain while the allocation of those benefits is uneven. Consequently, a proper evaluation distinguishes the total increase in purchasing opportunities froin the personal gains

realised by different households.

3.4 Distribution of Gains with Heterogeneous Endowments

Individuals and groups possess different endowments of labour, land or capital. Trade changes goods prices and can change the returns to these factors, raising incomes for some groups and lowering others. The analysis distinguishes aggregate gains from trade from their distribution across people with different resource ownership. Policy debate therefore includes not only whether trade expands the economy's opportunities but also which households, workers or asset owners gain.

Assume also that X is labor-intensive and that Y is capital intensive. Budget line for laborers and capitals are shown in Fig. 5.10, with each group's initial autarky income constraints given by AA' and each group's initial welfare level by Ua. Now assume that trade raises the price of X as in Fig. 5.1 from pa to p* . The economy responds to this change by shifting resources out of Y production and into X production. The output of the economy move from point A in Fig. 5.1 to point Q2. The commodity price changes caused by trade in turn cause factor price changes.

Unit Quick Revision

Exchange gains: better terms of consuming output. Specialisation gains: better allocation of resources. Aggregate gains and distribution of gains are different questions.

UNIT 3: CAUSES AND CONSEQUENCES OF TRADE

1.0 Introduction

To understand why nations trade, begin with a theoretical situation in which they have no reason to do so. Then remove the assumptions one by one. Differences in resources, technologies, preferences and policies create both trading opportunities and uneven effects.

3.1 The No-Trade Model

The no-trade benchmark assumes countries have identical technology and factor endowment proportions, similar homogeneous tastes, constant returns to scale, and no policy or market distortions. If these conditions imply the same autarky relative prices in each country, neither has an incentive to exchange at different prices. Five closely related conditions guarantee no trade: identical relative factor endowments, identical production technologies, constant returns to scale, identical homogeneous tastes and absence of distortions. The realistic differences from this benchmark help explain actual trade.

In fact, the trade of any country is a complex outcome of many causes all operating at the same time. A convenient method of examining the causes of trade is to first imagine a world in which there is no trade. What assumptions are necessary to ensure that the demand and supply situation in all countries are identical? On the demand side it is sufficient to assume that identical and homogeneous tastes exist throughout the world. The absence of distortions (e.g., taxes, subsidies, imperfect competition) While these five conditions are sufficient to imply that there will be no trade, there are obviously many other models that could be invented in which autarky prices would be identical so that no trade would take place.

PAST QUESTION - 2022_2 Q5(a)

Question: List five conditions that can guarantee a no-trade situation.

Answer: (1) Countries possess the same relative factor endowments; (2) they have identical technologies for producing each good; (3) production shows constant returns to scale; (4) tastes or preferences are identical and homogeneous; and (5) there are no distortions such as taxes, subsidies or imperfect competition. Together these conditions can yield identical relative autarky prices so that no gain arises from exchanging the two goods.

3.2 Methodological Considerations

Trade models deliberately simplify complex economies to identify causal relationships. Two-good and two-factor assumptions permit clear analysis of relative prices, resource allocation and gains. Results need qualification when transport costs, taxes, imperfect competition, unequal ownership of factors, varying tastes and factor mobility are admitted. The comparison of autarky and trade should distinguish positive analysis (what occurs) from welfare assessment (whether society or individual groups benefit).

To assess the relevance of this criticism, we must understand why this approach is being employed and what kinds of conclusions we expect to draw froin the analysis. It is clear that no conclusions about a specific cause of trade can be derived unless we can be sure that no other things are causing trade at the time. For example, we could not identify the effects of demand differences in a model in which endowments were also different, for it would generally be impossible to separate the effects of these two variables. At this stage of the analysis, then, the question of whether the model is “realistic” is not a relevant one, for no claim has been made about its predictive powers.

3.3 Consequences or Effects of International Trade

Major concerns include problems for developing countries: dual economies where export-oriented modern sectors coexist with neglected traditional ones; situations in which poorer nations capture limited benefits; possible deterioration in terms of trade for primary products; and demonstration effects where imported consumption standards increase demand for luxury items. Changes in global demand, substitution of synthetic products for agricultural goods and foreign protection of domestic producers can further constrain export incomes. Positive effects include access to goods, technology and larger markets, but distribution and resilience matter.

The effects of foreign factor movements have been that of creating a highly unbalanced structure of production of these countries. Although export increased but they did not contribute much to the development of the rest of the economy. During depression, terms of trade become adverse and their foreign exchange earnings fall steeply. It has been maintained that the inflow of foreign capital and developed a country’s natural resources only for export purposes, to the neglect of production in the domestic sector. Singer is also of the opinion that the benefits of technological progress have gone disproportionately to the advanced countries.

PAST QUESTION - 2024_1 Q3(a); 2022_1 Q5

Question: Explain the concept of dual economies as a consequence of trade.

Answer: A dual economy contains distinct sectors operating with different technologies and opportunities. In some economies, an export-oriented or modern sector existing alongside a less-developed traditional sector. The expanding trade sector may enjoy capital investment, equipment and market access while the other remains relatively neglected. Thus increased trade statistics do not by themselves establish that improvements are widely shared across the economy.

PAST QUESTION - 2021_2 Q5(b)

Question: Explain the adverse demonstration effect associated with international trade.

Answer: The demonstration effect occurs when exposure to foreign living standards and imported consumer goods stimulates imitation of consumption patterns that local income or production cannot comfortably sustain. Where scarce foreign exchange and saving are diverted toward luxury imports, investment in productive capacity may suffer. This is one possible adverse consequence for developing countries, not an inevitable outcome of trade.

3.4 Theoretical Issues in International Factor Movements

International economic relations involve movement of factors as well as goods. Capital may flow to foreign production and workers may migrate, even though trade models often assume limited international labour movement. Factor movements influence wages, production and the division of national income, and can change the quantities countries can produce. International factor mobility is treated as related to, but analytically distinct from, the exchange of finished goods.

The principles of international factor movements do not differ from those underlying international trade in goods. But international borrowing and lending, and international labour migration are similar in their causes and effects to the movement of goods. Nevertheless, there are major differences in the political context, (Krugman and Obstfeld, 2007). A labour abundant country may under some circumstances import capital intensive goods. A country that is too small to support firms of efficient size may import goods where large firms have an advantage to allow those goods to be produced locally by subsidiaries of foreign firms.

PAST QUESTION - 2025_1 Q2(b); 2024_1 Q4(b)

Question: Distinguish foreign trade from domestic trade and describe effects on poorer countries.

Answer: Foreign trade crosses national jurisdictions, so it encounters tariffs, import quotas and national policy controls that normally do not apply within a single country. Labour and sometimes capital are less mobile across borders; different currencies and institutional arrangements also matter. Developing countries may gain markets and access technology, but can face dependence, dual-sector development and unequal gains when export products meet weak demand or unfavourable terms of trade.

Unit Quick Revision

No-trade benchmark: similar endowments, technology, scale, tastes and no distortions. Actual trade produces both benefits and possible structural problems in less-developed economies.

MODULE THREE: DIFFERENCES IN TECHNOLOGY

UNIT 1: MODEL OF PRODUCTION FUNCTION DIFFERENCES

1.0 Introduction

The Ricardian approach isolates technological differences across countries as a source of comparative advantage. Labour productivity differs across industries and countries, creating relative unit labour requirements and domestic opportunity costs.

3.1 Model of Production Function Differences

A simplified Ricardian model uses labour as the productive factor and assumes constant returns to scale. Let ax and ay be units of labour needed to produce one unit of X and Y. A country has greater productivity where its unit labour requirement is smaller. Its opportunity cost for X in terms of Y is ax/ay. When countries differ in these relative input requirements, their autarky price ratios can differ and specialisation becomes profitable. Money wages help determine unit costs, but comparing nominal wages alone is not enough.

By differences in technology, we mean that the amount of output that can be obtained from one unit of labour differs across countries. The one-factor model can be thought of as a special case with one factor, the issue of differences in relative endowments does not arise. In terms of this production functions of one-factor model with constant returns will have a linear production possibility frontier.

3.2 Absolute and Comparative Advantage

Absolute advantage compares an industry’s physical efficiency across countries. Comparative advantage compares relative efficiency across goods. If Home requires fewer labour hours for both commodities, it has two absolute advantages, but it can have comparative advantage in only the good for which its relative efficiency is greater. A central point is that the opportunity-cost criterion governs trade even where one country is more productive in both sectors.

Note that α and β are the marginal products of labor in industries X and Y respectively: α and β give the additional outputs obtained from one unit of- labor. The Ricardian approach is illustrated in Table 7.1 (similar to Table 5.1), where we show the outputs of X and Y produced from one unit of labor in two countries, H and F. It is assumed that in Country II, 20 units of X are produced from one unit of labor, whereas 30 X can be produced in Country F with one unit of labor.

PSALMDATA

PAST QUESTION - 2023_2 Q1(a); 2025_1 Q3(a); 2025_2 Q4(a)

Question: State and discuss absolute and comparative advantage.

Answer: A producer has absolute advantage if it can produce a commodity with fewer inputs than another producer. Comparative advantage exists when its opportunity cost is lower. For example, if producing a unit of X costs Home one-half unit of Y but Foreign two units of Y, Home has comparative advantage in X, whether or not it is absolutely more productive. Trade takes place because exchanging at a price between these domestic relative costs benefits both partners.

3.3 The Production Possibility Frontier

With only labour and fixed labour requirements, full employment gives axX + ayY = L, a straight-line PPF with intercepts L/ax and L/ay. Its slope is -ax/ay. A lower relative unit labour requirement for X makes that country relatively efficient at X. The slope therefore connects technological differences to autarky relative prices and the direction in which each country will tend to export.

The distance of a country's production frontier froin the origin depends the absolute levels of its labor production coefficients (α, β) its labor endowment (L). The slope of a country's production frontier is simply the ratio of that country's labor productivity coefficients: - □Y/□X = β/α. In autarky, Home and Foreign are in equilibrium not points Ah, and Af, respectively. This simple Ricardian model with linear production frontiers has the property that a country's autarky price ratio is given by the slope of its production frontier.-Because the slope of the production frontier reflects a country's comparative O H F X P1 > Pha H C2 C1 Ah O H X advantage, autarky prices reflect comparative advantage.

PAST QUESTION - 2022_1 Q3

Question: Explain how a Ricardian PPF reflects differences in technology.

Answer: In the one-factor model, fixed labour requirements satisfy axX + ayY = L. Setting X to zero gives maximum Y = L/ay; setting Y to zero gives maximum X = L/ax. Joining these intercepts produces a straight PPF with slope -ax/ay. Countries with different ax/ay ratios have different transformation costs and hence different comparative advantages. If both sectors can trade at more attractive world prices, the resulting exchange can raise welfare.

3.4 Excess Demand and International Equilibrium

Trade opens when countries face world relative prices different from their autarky ratios. Their production shifts toward the sectors in which they have comparative advantage, while consumers adjust to the world exchange rate. International equilibrium requires one country’s net exports to match the other’s imports. The common price ratio must be consistent with labour allocations, world demand and trade-balance conditions.

In the Ricardian model, autarky prices correspond to relative unit labour requirements. A world price ratio lying between the two countries’ autarky ratios can make production of one good attractive in Home and production of the other attractive in Foreign. The precise output and consumption positions depend on preferences and labour supplies, but the central condition is that the value of imports be balanced by exports. If world prices were identical to each country’s pre-trade relative prices, the incentive to move toward specialisation would disappear. Thus

differences in technology produce trade through differences in relative costs.

Unit Quick Revision

Ricardian model: unit labour requirements reveal opportunity cost. Absolute advantage is higher productivity; comparative advantage is lower relative cost. With fixed coefficients, axX + ayY = L.

UNIT 2: ROLE OF WAGES IN THE RICARDIAN FRAMEWORK

1.0 Introduction

Wages are both incomes earned by labour and part of production costs. In the Ricardian model, productivity and wage rates interact: a country paying higher wages can still export successfully when its productivity advantage outweighs the wage difference.

3.1 Wage Theory

Economic thought includes wage theories to show that wages have been explained from several viewpoints. Subsistence approaches relate wages to the cost of maintaining labour. Marginal-productivity reasoning links remuneration to the additional output labour produces. Surplus-value approaches consider the distribution of output between labour and owners; the residual-claimant theory views wages as what remains after other factors receive their shares; and bargaining theory emphasises the relative negotiating power of employers and employees. Historical theories must be understood in their historical settings.

Classical theories The Scottish economist and philosopher Adam Smith, in The Wealth of Nations , failed to propose a definitive theory of wages, but he anticipated several theories that were developed by others. Moreover, he noted that workers would need to be compensated by increased wages if they were to bear the cost of acquiring new skills—an assumption that still applies in contemporary human-capital theory. The English classical economists who succeeded Smith, such as David Ricardo and Thomas Malthus, held a more pessimistic outlook. Smith defined this theoretical fund as the surplus or disposable income that could be used by the wealthy to employ others.

PAST QUESTION - 2023_2 Q1(b); 2021_2 Q1(b); 2022_2 Q2(b)

Question: Outline major theories of wages.

Answer: Several theories offer different explanations of wage determination, including the subsistence theory, wage-related fund approaches, the marginal-productivity view, the surplus-value theory, residual-claimant theory and bargaining theory. The subsistence approach links wages to labour maintenance; marginal productivity connects wages to labour's contribution to output; the surplus-value approach examines how the product is divided; residual-claimant reasoning treats wages as the remaining share; and bargaining theory links the settlement to the negotiating strength of employers and workers.

PAST QUESTION - 2025_2 Q3(b); 2022_2 Q5(b)

Question: Differentiate residual-claimant and bargaining theories of wages.

Answer: Residual-claimant theory treats wages as the portion of product value left after land, capital and other claimants receive their rewards. It therefore makes the wage bill a residual in the distribution process. Bargaining theory states that actual wages, working hours and employment conditions are determined through negotiation and the relative bargaining power of employees and employers. The first is chiefly a theory of distributive shares; the second stresses bargaining institutions and strength.

3.2 The Role of Wages

When productivity differs between countries, nominal wage differences do not determine comparative advantage by themselves. In competitive equilibrium, a firm can pay higher wages if each worker produces more output. The relevant international cost of a good is the wage rate multiplied by unit labour requirements. Comparing those costs across industries determines which country can supply each good competitively. Trade can change the purchasing power of wages as workers use income to buy domestic and imported goods.

3.3 Model of Wages in the Ricardian Framework

Let wH and wF be wages in Home and Foreign, and aHX and aFX the respective unit labour requirements in commodity X. Unit labour costs are wH aHX and wF aFX. Home can supply X at lower cost if wH aHX < wF aFX. Similarly compare costs for Y. The model connects productivity with international wage gaps and shows why an absolute productivity advantage can correspond to a higher real wage while the direction of trade follows comparative cost differences.

Relative prices must equal relative real costs in terms of labor inputs. Here, if sector Y has a relatively low marginal product and sector X has a relatively high marginal product, the home country is likely to have its comparative advantage in good X. The wage rate, y which is the nominal price of labor, has no effect on relative commodity prices as long as it is the same in both sectors. The wage rate is relevant for determining real wages, or the living standards of laborers. This change will alter nominal wages in both countries, generating wage rates Wh and Wf.

PAST QUESTION - 2024_1 Q5(b)

Question: Derive the Ricardian wage-cost comparison.

Answer: Let wage rates in two countries be wH and wF, and units of labour required per unit of X be aHX and aFX. Unit costs are wH aHX and wF aFX. Home has a cost advantage for X if wH aHX < wF aFX, or wH/wF < aFX/aHX. Apply the corresponding inequality to Y to establish the feasible wage range compatible with each country specialising according to comparative advantage. Productivity differences, not just nominal wages, determine competitive supply.

Unit Quick Revision

Wages must be compared with productivity, not in isolation. Competitive unit cost = w × labour hours per unit. Residual-claimant theory differs from bargaining theory.

UNIT 3: THE HECKSCHER-OHLIN MODEL

1.0 Introduction

The Heckscher-Ohlin model shifts attention from differences in technology to differences in countries' factor endowments. Countries with relatively abundant labour and capital face different relative production opportunities, even with identical production methods.

3.1 Effects of Endowment Differences

The model commonly assumes the same production technologies and preferences across trading countries, constant returns to scale and competitive markets. If Home has relatively more capital per worker than Foreign, Home's production frontier is biased toward the capital-intensive commodity. Relative endowments shift relative costs and may produce different autarky prices. Opening trade creates a reason to export the abundant-factor-intensive good and import the scarce-factor-intensive good.

These production functions, which are the same in both countries, differ in relative usage of capital and labor. Second, there are fixed total supplies of the two factors, labor and capital, which are homogeneous and perfectly mobile between industries within each country. However, labor and capital arc assumed to be perfectly immobile between countries. Third, there are no market distortions such as imperfect competition, labor unions, or taxes that would influence production or consumption decisions. When expanding the model to allow for trade, two additional assumptions are required. First, preferences in both countries are taken to be identical and homogeneous.

3.2 Factor Endowments

A factor endowment is the amount or availability of a productive resource owned or usable by a country, such as labour and capital. Relative rather than simply absolute abundance matters. A nation can have a large workforce and still be relatively capital-abundant if its capital-to-labour ratio exceeds that of its trading partner. The economic value of endowments is revealed partly in relative factor prices, such as wages compared with capital rentals.

First, note that it is relative endowments that affect the measurement of factor abundance for example, although the United States has a larger labor force than either Brazil or Mexico, it is capital-abundant and labor-scarce because it has a comparatively larger capital supply. Switzerland has the smallest absolute capital stock, hut it is the most capital-abundant country in the group. Second, while the Heckscher-Ohlin theory assumes these factor endowments to be fixed, it should be clear that capital and labor supplies depend to some degree on economic conditions. For example, the size of the aggregate labor force may well depend on the wage rate and on the attractiveness not working conditions.

3.3 Factor Intensities

Factor intensity describes how a sector combines inputs. X is labour-intensive relative to Y if it uses a lower capital-to-labour ratio, $Kx/Lx < Ky/Ly$, at the relevant factor prices. Endowments characterise countries; intensities characterise commodities and industries. A country abundant in labour is predicted to specialise relatively in the labour-intensive good.

In principle, it is possible that at different relative factor prices, the rankings in Eq, can be reversed if one industry finds it technically easier to substitute capital for labour along an

isoquant than does the other. For this point, the maximum producible quantities of X and Y are X and Y. Now consider the effect of changing the endowment point for one of the, two countries. Here, the increase' in the endowment of capital and the reduction in the endowment of labor result in a rise in the maximum output of Y, the capital-intensive commodity, and a reduction in the maximum output of X, the labor-intensive commodity.

PAST QUESTION - 2025_1 Q3(b); 2021_2 Q5(c)

Question: Distinguish factor endowments from factor intensities.

Answer: Factor endowments describe a country's resource stocks or relative abundance, such as its national K/L ratio. Factor intensities describe the mix of inputs used in particular goods, such as Kx/Lx for X and Ky/Ly for Y. Endowment comparisons are country-to-country; intensity comparisons are industry-to-industry. The Heckscher-Ohlin theorem links the two by predicting export of the product intensive in a country's abundant factor.

3.4 The Heckscher-Ohlin Theorem

The Heckscher-Ohlin proposition is that countries export commodities using relatively abundant factors intensively and import commodities using relatively scarce factors intensively. If Home is capital-abundant and X is capital-intensive, Home tends to export X. The standard result relies on common technology, constant returns, competitive conditions and a meaningful factor-intensity ranking. With trade, the resulting commodity-price changes also affect relative wages and returns to capital, connecting this unit with Stolper-Samuelson and factor-price equalisation.

An interesting question regarding the Heckscher-Ohlin theorem relates to how far we can relax the underlying assumption find still ensure that the result holds. If factors were not homogeneous, meaning that labor was distinguished by skills and that capital came in different types, the simple two-factor theorem would no longer be relevant. If factors are not mobile between industries but instead must remain lived in employment for some period of time, the model must depart froin the long-run nature of the Heckscher-Ohlin theorem in favor of a short-run view. However, if factors could migrate internationally, they would do so to take advantage of differences in factor returns.

PAST QUESTION - 2023_2 Q2(a,b); 2021_2 Q2(a,b); 2025_2 Q5(a)

Question: Explain the Heckscher-Ohlin model and the effect of factor endowment differences.

Answer: The model explains trade through differences in relative stocks of factors of production. Countries are assumed to possess access to the same technology while goods use factor inputs with different intensities. A relatively labour-abundant economy can make labour-intensive products more cheaply in relative terms and tends to export them. A capital-abundant economy tends to export capital-intensive products. Endowment differences bias the production possibilities and produce relative price differences, creating opportunities for mutually beneficial trade.

PAST QUESTION - 2025_2 Q4(b); 2021_2 Q2(c)

Question: Mention three production-related assumptions of the Heckscher-Ohlin model.

Answer: (1) The same production technologies are available in the two trading countries. (2) Production functions exhibit constant returns to scale. (3) Commodities have stable and distinct factor intensities, so one good can be regarded as relatively labour-intensive and another as relatively capital-intensive. The wider model also uses competitive markets and consistent preferences in its standard presentation.

Unit Quick Revision

Endowments describe countries; factor intensities describe industries. Heckscher-Ohlin: a nation exports the good using its relatively abundant factor more intensively.

UNIT 4: THE STOLPER-SAMUELSON AND RYBCZYNSKI THEOREMS

1.0 Introduction

The two major results of this unit explain different consequences of trade. Stolper-Samuelson connects changes in commodity prices to real factor incomes; Rybczynski connects changes in factor endowments to the output pattern when product prices remain fixed.

3.1 The Stolper-Samuelson Theorem

In a competitive two-good, two-factor economy, an increase in the relative price of a good raises the real return to the factor used intensively in its production and lowers the real return to the other factor, under the stated assumptions. If labour-intensive X becomes relatively dearer, real wages rise and returns to capital fall. The mechanism is the reallocation of production and the resulting factor demand. The theorem shows why the nation can gain from trade while some owners of factors lose.

However, there is an even stronger theorem, which was first proven by two American economists, Wolfgang Stolper and Paul Samuelson. Of course, given fixed factor endowments, this is possible only because output of X rises and output of y falls. The movement from A to D involves a reduction in the amount of labor used in the production of X. while the input of capital remains constant. Now consider the vertical movement from ,A to E in Fig. 8.7, which represents an increase in the use of capita) in industry X with an unchanged labor input.

PAST QUESTION - 2023_2 Q3; 2025_2 Q5(b); 2024_2 Q5(b)

Question: Discuss the Stolper-Samuelson theorem.

Answer: The theorem states that, given its competitive two-factor assumptions and both goods continuing to be produced, a rise in the relative price of a commodity raises the real reward to the factor used intensively in that commodity and reduces the real reward to the other factor. Thus a price rise for a labour-intensive good raises real wages but lowers real capital returns. It follows that trade-induced changes in goods prices affect the distribution of national income, so total gains from trade need not mean gains to every group.

3.2 The Rybczynski Theorem

Holding relative commodity prices constant, if one factor endowment expands and both commodities continue to be produced, output of the commodity using that factor intensively rises while output of the other good falls. For example, additional labour increases the labour-intensive sector's output and draws resources from the capital-intensive sector. An important aspect is the magnification effect: the output shift can be more than proportional because the expanding sector absorbs not only new workers but other factors reallocated from the contracting sector.

This can lead to an adjustment in the quantities and types of outputs between the two regions. Recall that point A represents the common tangency between isoquants from both industries. The effects of the increase in the labor supply on outputs art easily seen. The Rybczynski theorem has important implications for the effects of changes in endowments on production possibility curves. After the increase in the labor supply, the new equilibrium position will be A', where there is a larger output of X but a smaller output of Y. The Rybezynski lines are it; fact linear if constant returns to scale exist in both industries.

PAST QUESTION - 2023_2 Q4(a)

Question: Explain the Rybczynski theorem.

Answer: At constant commodity prices, an increase in the supply of one factor increases production of the commodity that uses it relatively intensively and reduces production of the other commodity, as long as both sectors operate. If labour expands, the labour-intensive sector grows and the capital-intensive sector contracts. Its logic rests on factor reallocation at unchanged prices, and the resulting changes can be magnified relative to the original increase in labour.

3.3 Trade Theory with Many Goods and Factors

Moving from two goods and two factors to many products complicates precise commodity-by-commodity predictions. In this setting, rankings by factor intensity no longer always yield a unique trade pattern. Nevertheless, a country rich in a factor is generally expected to export a bundle of commodities embodying relatively more of that factor. The reasoning shifts from an individual exported commodity to the aggregate factor content of the entire trade bundle.

When products are numerous, factor intensities form more than one simple ranking. A country might export some relatively labour-intensive products and also some sophisticated capital-intensive goods, depending on their specific costs and the wider production structure. It can still be useful to compare total factor content of all exports with total factor content of all imports. The many-goods extension therefore makes a narrower prediction about individual commodities but retains the basic link between endowments and the factor composition of a country's trade bundle.

3.4 The Factor-Content Theorem

Factor-content analysis examines the services of productive factors embodied in exports and imports, rather than comparing goods only by industry. A capital-abundant country is expected to be a net exporter of capital services embodied in its merchandise trade, while a labour-abundant country exports embodied labour services. This formulation makes it possible to discuss trade patterns with many traded goods and helps connect observable factor use with the Heckscher-Ohlin proposition.

Its practical importance is in showing that, under reasonably general circumstances, differences in relative endowments still determine comparative advantage. However, comparative advantage in this sense refers to the pattern of trade in factor services rather than in goods. Indeed, it does not really matter that we cannot predict whether a country will export or import a particular commodity, because commodity trade is merely the means for implicitly trading factors. The factor-price-equalization theorem also generalizes for any number of factors and goods as long as the number of goods is equal to the number of factors and all countries produce all these goods (that is, they are incompletely specialized").

PAST QUESTION - 2023_2 Q4(b); 2024_1 Q4(a); 2021_2 Q3(b)

Question: Discuss the factor-content theorem.

Answer: The factor-content theorem examines the labour and capital services embodied in a country's exports compared with those embodied in imports. It generalises the endowment argument: a capital-abundant country should, under model assumptions, export a bundle using relatively more capital, while a labour-abundant country exports a bundle using relatively more labour. The result refers to the total factor composition of trade, not simply to the label attached to an individual exported product.

PAST QUESTION - 2023_2 Q5(b)

Question: What is factor-price equalisation?

Answer: Factor-price equalisation is the prediction that free trade in goods can, under restrictive common-technology, competition, diversified-production and other assumptions, bring relative factor rewards such as wages and returns to capital into alignment across countries. Commodity-price equalisation affects the demand for factors used to produce them. It does not mean that every country must display identical wage levels in actual imperfect economies.

3.5 International Product Cycle Theory

Products may move through innovation, expansion and standardisation stages, with the location of production and trade changing as technology spreads and manufacturing becomes more routine. Product-cycle reasoning can be considered alongside factor-content theory to show that trading patterns need not remain fixed over time. As new production techniques become familiar, comparative advantage can shift toward countries able to manufacture established goods at lower cost.

These countries possess abundant capital and research and development (R & D) capabilities. Here, each product and its associated manufacturing technologies go through three stages. The product's investors later produced the good and exports it to foreign markets. As the producer's manufacturing becomes more standardized, foreign competitors enter the market place and the

monopoly powers of the investors dissipates. In fact, foreign competitors may enjoy a comparative advantage in producing the matured product. This will have the effect to fulfill the needs of exports markets as well Product standardization will eventually progress. For instance, early in the product's evolution manufacturing requires highly-skilled knowledge of workers in R&D.

Unit Quick Revision

Stolper-Samuelson: goods-price change redistributes real factor income. Rybczynski: more of one factor raises its intensive good output and lowers the other at constant prices. Factor content concerns inputs embodied in trade.

MODULE FOUR: THE FACTORS MODEL AND GOVERNMENT POLICIES AS DETERMINANTS OF TRADE

UNIT 1: THE SPECIFIC FACTORS MODEL

1.0 Introduction

The specific-factors model separates mobile labour from factors tied to particular industries over the relevant time period. It helps explain how a change in a product's price can alter wages and the returns earned by sector-specific capital or land.

3.1 Productivity of Labour

Labour productivity is output per unit of labour, while marginal productivity measures additional output obtained from adding an extra worker with other resources fixed. In a model where labour can move between industries but some factors cannot, employing more labour in one sector normally reduces its marginal product. Labour moves in response to available wages until no worker can earn more by transferring between sectors.

Because some capital is tied to a particular industry, the number of workers employed there affects labour productivity. If that industry attracts more labour without adding machinery, each additional worker normally contributes less extra output. Its marginal product curve therefore slopes down over the relevant range. When the wage offered by one sector rises above the other, mobile labour transfers until wage opportunities again balance. The owners of immobile or sector-specific equipment cannot respond as quickly; for that reason their income changes may be greater or more direct than those experienced by workers who can move.

3.2 Determination of Marginal Productivity

For sector X, marginal product of labour is MPLx = change in X divided by change in Lx at fixed specific capital. Under competition, the value of marginal product is the commodity price times MPLx. Firms hire labour to the point at which w = Px MPLx; similarly w = Py MPLy in sector Y. The common wage therefore helps allocate mobile labour between competing sectors, whereas returns to industry-specific factors adjust differently when prices change.

Note that froin these conditions we can specify that the real returns to labor and capital in terms of commodity X are equivalent to their marginal products: wfpx = MP/jf and r/px = MPx, respectively. This result means that changes in real returns to factors are known as soon as changes in capital-labor ratios are determined. For example, in Fig. 9.1, a rise in the capital

stock for a given level of employment generates a higher real wage. Moreover, because marginal products are functions of only the capital-labor ratios, if we know how one marginal product changes, we immediately know that the other marginal product has changed in the opposite direction.

3.3 Commodity Prices and Factor Prices

A rise in the price of one commodity raises the value of marginal product of labour in that sector and draws workers toward it. The wage can rise in money terms, but the gain in the real wage depends on which consumer good is used as the deflator. The owners of the factor specific to the expanding sector generally gain, while the owners of the other specific factor may lose. An important point is that who gains depends on factor mobility and the identity of the sectors.

Suppose the international price of good X rises while good Y has an unchanged price. Higher output value in X raises its demand for labour and attracts workers from Y. The money wage increases, but less than the increase in the price of X in the usual model. Workers may therefore be able to buy more of Y while their wage measured in units of X falls. Owners of the factor specific to X gain from stronger demand, whereas owners of the factor specific to Y face pressure. The example shows why the effects of trade cannot be judged by money wages alone.

PAST QUESTION - 2023_2 Q5(a); 2025_2 Q5(a)

Question: Differentiate commodity prices from factor prices.

Answer: A commodity price is the money or relative price paid for an output good, such as X or Y. A factor price is remuneration for an input used in production, such as the wage for labour or rental for capital. Product prices affect the value of marginal productivity; in a competitive sector, employment decisions are related by w = P times marginal product of labour. A rise in an export good's price can therefore shift wages and the earnings of industry-specific factors.

3.4 Exchange Rate Policies

The main categories are freely floating exchange rates (market demand and supply largely determine the rate), managed floats (authorities intervene while permitting movements) and fixed or pegged systems (the rate is officially linked to an anchor). Flexible rates permit adjustment to economic changes but entail currency risk; fixed rates improve rate predictability but may require substantial reserve and policy commitments. Exchange-rate policy affects the domestic price of traded commodities, import bills and export receipts. The unit also lists payment arrangements used in international trade.

Under a floating system, market forces are permitted to alter currency prices when demand and supply change. A managed float allows official intervention to limit or guide movements. A fixed-rate system promises stability against an anchor but may require reserves or direct controls when market pressures increase. For exporters, exchange-rate depreciation can raise domestic-currency receipts from foreign sales; for importers, it can make overseas purchases and foreign-currency debts more costly. These effects depend on prices and contracts, so the distinction between short-term adjustment and longer-term production responses is important.

Unit Quick Revision

Specific factors cannot move freely across sectors over the chosen period. w = P × MPL in competitive labour markets. Product-price and exchange-rate movements affect workers and specific-factor owners differently.

UNIT 2: ENDOWMENT CHANGES AND FACTOR PRICES

1.0 Introduction

Endowments change as labour forces grow, capital is accumulated, or resources move across borders. The resulting changes in output and factor earnings depend on which factors are mobile, which are tied to particular sectors and how prices adjust.

3.1 Endowment Changes, Factor Prices, and Outputs

Adding workers expands potential production but can reduce the marginal product of labour when the stock of specific capital is held fixed. A useful comparison concerns the effects of larger labour and specific-factor endowments. A rise in one industry's specific capital raises its demand for mobile workers; workers are attracted from the other industry until wages adjust. The returns to various factors do not move identically because their supplies and uses differ.

The existence of different factor endowments does not change this result as long as countries are incompletely specialized. In this context it is no surprise that factor price equalization, ice; lot hold in the specific-factors model, because it embodies three factors and only two goods. If exports good X, then the real income of capital will rise and that of S-capital will fall, as we demonstrated. There is clearly no presumption toward equalization of real factor prices in the short run. Consider the effects of endowment changes on real incomes, as depicted in Fig. 9.3, where we take as the initial situation the free trade equilibrium at point C.

3.2 Factor Endowments and Factor Prices

A country's factor endowment includes its labour and capital supplies; factor prices are the wage and returns to other productive assets. The relationship depends on technological conditions and ability to substitute factors. In a two-sector specific-factors setting, additional labour can raise both sectors' total output but reduce the real wage relative to specific capital owners. An increase in capital specific to one sector can attract workers and change the real returns earned in both sectors.

Again, to make the contrast to the factor-price-equalization theorem, recall that, in that model, any endowment change resulted in no impacts on the real returns to any factor as long as the economy remained incompletely specialized. In the specific-factors model, however, endowment shifts have sharp impacts on real factor incomes, whether or not the economy is specialized. To make the comparison to the factor-price-equalization theorem fully explicit, the rise in the endowment of the specific factor S in the small open economy of Fig. 9.3 cannot have an impact on commodity prices or factor prices in the rest of the world.

PAST QUESTION - 2023_2 Q5(b)

Question: Summarise factor-price equalisation and its limitations.

Answer: Trade can bring commodity prices closer together, which in the standard diversified, common-technology model tends to equalise the returns to labour and capital across countries. That result requires strong assumptions about production and competition. Where specific factors, different technologies, trade barriers or incomplete specialisation operate, equal prices for traded goods need not imply identical wages and returns to capital.

3.3 Endowment Changes and Outputs

Changing the supply of one factor shifts the economy's output possibilities toward activities that use the expanding resource more heavily. At fixed product prices in a standard Heckscher-Ohlin setting, the Rybczynski effect predicts expansion of the intensive sector and contraction of the other sector. In the specific-factors setting, effects differ because some productive resources cannot leave their sectors quickly. The model chosen determines the precise direction of adjustment.

Again, this is due to the mobility of labor rather than to the intensity of production. Factor endowments and outputs An increase in one specific factor increases the output of the commodity that uses that factor and reduces the output of the other industry. Increases in the supply of the mobile factor will expand both outputs. It is clear from our discussion that mobile factors would oppose policies allowing freer immigration of competing mobile factors, while owners of specific factors would favor them. Again, this is a result of factor mobility rather than factor scarcity, as in the Heckscher-Ohlin model.

3.4 The Pattern of Trade

In the specific-factors model, trade patterns depend on the ownership and size of sector-specific resources as well as relative labour availability. An economy may export goods using its relatively large specific-factor endowment. Endowment growth can alter comparative costs and hence future export composition. The analysis points out that factor-specific circumstances are especially relevant to short-run trade and policy debates.

A larger capital stock or a bigger workforce changes the potential scale and composition of production. If the extra capital belongs to export-oriented manufacturing, output in that sector may expand and change the country's comparative costs. In another case, growth in labour supply can alter relative factor returns and shift production toward more labour-using activities. Relative world prices also matter: a change in domestic supplies produces a different market response when the country is too small to influence world prices than when it can alter its terms of trade.

PAST QUESTION - 2025_2 Q5(a); 2023_2 Q2(a)

Question: Explain factor endowments and their effect on a country's trade pattern.

Answer: Endowments are the available stocks of labour, capital and other productive inputs. Differences in relative factor supplies affect the goods a country can produce relatively cheaply. In the Heckscher-Ohlin case, a country exports the commodity intensive in its abundant factor. When endowments change through investment, population growth or migration, relative production opportunities may shift and the composition of exports and imports may change. The analysis also considers short-run effects when factors are tied to individual industries.

Unit Quick Revision

Changes in labour, capital and sector-specific resources alter output and returns. The specific-factors model and the Heckscher-Ohlin model have different adjustment assumptions.

UNIT 3: GOVERNMENT POLICIES AS DETERMINANTS OF TRADE

1.0 Introduction

Governments change the incentives to produce, purchase and trade by altering prices through tariffs, taxes or subsidies. Trade patterns created by such measures are not automatically welfare-improving, so it is necessary to distinguish the price consumers see, the price producers receive and the world price.

3.1 Consumer, Producer, and World Prices

Without distortions, producers and consumers face the same relative prices, which under free trade equal world prices. A consumption tax makes the price paid by buyers diverge from the world price and can discourage consumption of the taxed commodity. A production tax or subsidy makes firms respond to a price different from the world price while consumers can still face international prices. Separate price lines show to show how these wedges shift the production or consumption point.

For analysis, denote the consumer price ratio by q, the producer ratio by p and the world price ratio by p*. With no distortion these may coincide. A tax on consumption introduces a difference between q and p*, leading households to alter their spending pattern even if producers continue to face world prices. A production tax or subsidy instead drives a wedge between p and p*, altering what firms find profitable to supply. Drawing separate price lines for firms and buyers helps explain why the effect of a policy on the trade balance may differ from its effect on welfare.

3.2 Taxes, Subsidies, and Trade Incentives

A tax on consuming Y encourages substitution toward X, while a tax on producing Y discourages its production. A subsidy to producers of X can artificially expand X output and change the goods exported even where the undistorted economy had little reason to trade. Each policy creates a wedge between market signals and true resource opportunity costs. In the standard diagrams, these distortions can reduce welfare relative to the undistorted benchmark even if exports increase.

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When there are only two goods, the effects of a tax on one good are equivalent to a subsidy on the other good. These latter assumptions are reflected in the fact the consumption and production bundles are the same even though, in the case of a tax, for example, the consumption bundle costs more than the value of those goods at producer prices. The consumers pay more than the producers receive because of the tax, but then they receive an income in excess of the value of production because they receive the tax refund. The first bracketed term on the right-hand side is income received from production (payments of factors of production), while the second term on the right-hand side is redistributed tax revenue.

3.3 Taxes and Subsidies as Determinants of Trade

Policy-induced trade can arise because producers and consumers react to different relative prices. When taxes or subsidies redirect production away from the highest-value global allocation, some resources are used inefficiently. Benefits of an intervention may still be discussed if there are separate external costs, such as pollution, that ordinary market prices ignore; however, the central proposition is that increasing exports through distortion alone is not proof of an economic gain.

Once we introduce trade, we have to keep track of not only consumer and producer prices, but also world prices. In the closed economy it does not matter, because production and consumption of each good are equal. With a consumption Lax, consumers pay a tax on both domestic and imported goods. The relationships among consumer, producer and world prices with a consumption tax on Y are given by The second equation emphasizes that producers in H face world prices. Producers will continue to produce at A where the world price ratio, equal to the producer price ratio, is tangent to the production frontier.

PAST QUESTION - 2024_1 Q5(a)

Question: How can government policy influence gains from trade?

Answer: Government measures such as taxes, subsidies, tariffs and import restrictions change producer and consumer relative prices. They may change what a country imports and exports or encourage a favoured sector to expand. However, the analysis shows that trade generated by an artificial price distortion need not raise welfare: an intervention can push production inside the efficient frontier or induce consumers to buy an inefficient bundle. Evaluate policy by its effect on overall welfare and any genuine market failure it addresses.

3.4 Evaluating Protection and Free Trade

Free trade permits production and consumption choices shaped by world prices, while protection introduces tariffs, quotas or other restrictions affecting the level and composition of imports. In some economies, how countries control trade and explains that welfare analysis must consider who bears the price wedge. Protection can support a domestic industry, but also raise consumer costs and create inefficiencies. The key economic test is the welfare and resource-allocation effect, not the numerical rise in one sector’s output.

PAST QUESTION - 2025_1 Q4(b,c,d)

Question: Distinguish free trade, protection, quotas and the infant-industry concern.

Answer: Free trade involves comparatively unrestricted exchange at international prices. Protection refers to government restrictions or price measures designed to limit foreign competition or encourage local production. A quota directly limits the permitted quantity of imports. It is recognised that international competition can threaten developing home industries, which is the concern behind protecting infant firms; nevertheless, it requires analysis of the costs imposed on consumers and of whether sheltered production becomes efficient.

Unit Quick Revision

Distinguish prices faced by consumers, producers and the world market. Taxes and subsidies create wedges. More exports caused by an intervention need not mean higher national welfare.

UNIT 4: GAINS FROM TRADE AS A RESULT OF GOVERNMENT POLICY

1.0 Introduction

The final unit revisits gains from trade when an economy is already affected by taxes or factor-market distortions. The familiar theorem must be used carefully: a larger volume of international trade is not sufficient evidence that welfare has improved.

3.1 Gains from Trade: A Formal Analysis

In the undistorted model, maximisation of production value at world prices and balanced trade support a wider consumption opportunity set. With a tax or subsidy, consumer and producer prices diverge from world prices, so comparing world-price values alone does not fully establish welfare. A formal approach reformulates the gains-from-trade inequality to include the distortion term. Whether trade improves welfare may depend on how the taxed or subsidised activity responds relative to the autarky position.

But in order to evaluate welfare, we need to have consumption at consumer prices, not producer prices. The rightmost term is positive if trade increases the value of Y consumption. A sufficient condition for gains from trade is that consumption of the tax-distorted good increases. To put the matter the other way around, a country may fail to gain from trade when trade further reduces the consumption of a taxed good (or increases consumption of a subsidized good). Comparing the undistorted autarky equilibrium A in Fig, 10.2 to distorted free trade consignation at C. the sufficient condition fails.

PAST QUESTION - 2024_1 Q5(a)

Question: Explain why government policy may not always generate real gains from trade.

Answer: Tariffs, taxes and subsidies alter the prices facing consumers and producers, so individual choices can depart from those that maximise social welfare. Even where the intervention increases exports of a targeted commodity, the country may sacrifice more output value elsewhere or distort household consumption. If factor markets are also distorted, production can take place inside the efficient PPF. The gains-from-trade argument must therefore incorporate the price wedge, not merely the amount of international trade.

3.2 Factor Market Distortion

A tax on capital used in one sector can cause that industry to face a different rental cost from other producers. The firms then operate with unequal marginal rates of technical substitution, so resources need not be efficiently allocated between sectors. Production may occur inside the efficient PPF, and the distorted feasible production boundary may not have the conventional concavity. The analysis illustrates why domestic factor-market distortions can change the welfare consequences of opening trade.

Consider capital rented to the X and Y industries. Without factor-market distortion, both industries can choose combinations of capital and labour based on the same rental-to-wage ratio, and an efficient allocation equates their marginal rates of technical substitution. A tax confined to capital used by one industry changes its effective rental cost. That sector then faces a different relative input price froin the other sector, preventing an equalisation of their technical substitution rates. An apparently profitable output mix for separate firms can therefore waste resources froin the viewpoint of the economy as a whole.

3.3 Competitive Equilibrium

Competitive equilibrium in the simple, undistorted benchmark links factor allocation, firm profit maximisation and consumer choice through prices. With distortions, the price signals facing firms or households can differ. Producers may no longer operate at the social maximum-value allocation, even if each individually maximises profit at its own distorted prices. Policy evaluation therefore requires checking efficiency in both consumption and production.

The distorted autarky equilibrium must be at a point like C in Fig. 10.7, where the indifference curve is flatter than the distorted production frontier. The difference here is that the non-'tangency is along the distorted production frontier. The factor market distortion thus involves two distortions: One distortion can be thought of as the movement inside the efficient production frontier and the other as the movement along the distorted frontier. trading equilibrium, and under the assumption that we begin with two identical countries. The initial production frontiers are given by TAT’ with autarky equilibrium at A for each of the two countries.

3.4 Balance of Payments

It is useful to distinguish the simple trade-balance condition used in the real trade model from broader international settlement questions. The two-good model requires imports to be paid for by exports at world prices. In practice, international accounts can also involve factor-income payments, capital flows and reserves; the model’s simplifying trade-balance equality is the appropriate starting point for the diagrams and formal welfare proof in this unit.

When all the elements are correctly included in the BOP, it should be zero in a perfect scenario. On the other hand, the BOP deficit indicates that its imports are more than its exports. All transactions will have a debit entry and a corresponding credit entry. Japan books the export of the 100 cars as a debit in the BOP, while the U.S. books the imports as a credit in the BOP. A country’s BOP is vital for the following reasons: The BOP of a country reveals its financial and economic status. The BOP statement helps the government to decide on fiscal and trade policies.

PAST QUESTION - 2021_2 Q3(a)

Question: Explain trade balance in the two-good model.

Answer: In the standard model the value of goods produced at world prices equals the value of consumption at those prices, because the country pays for imports with exports: Px*Xp + Py*Yp = Px*Xc + Py*Yc. Equivalently, the weighted excess demands add to zero. This condition allows domestic output and consumption to differ good by good while the value of exports offsets the value of imports.

Unit Quick Revision

A distorted economy needs a modified gains-from-trade argument. Tax wedges and factor-market distortions can make resources inefficiently allocated and trade benefits uncertain.

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